

Characterizing a Low-Dimensional Redshift-Dependent Difference Between CMB- and Distance-Compatible Expansion Histories

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Abstract

Under the endpoint constructions examined here, low-redshift distance data and Planck-calibrated references favor substantially different best-fit expansion histories: neither side fits the full DESI DR2, Pantheon+SH0ES, and Planck TT/TE/EE likelihood contract at once, and none of the tested endpoints simultaneously supplies a complete fit to all sectors. We compare three frozen endpoints over the constrained domain $z \leq 1.8$: a phenomenological low-redshift distance-sector endpoint R reaching $H_0 = 72.80 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (not presented as a complete cosmological model: no recombination, BBN, or perturbation completion is claimed), a Planck-calibrated reference C1 with $H_0 = 67.99 \text{ km s}^{-1} \text{ Mpc}^{-1}$, and a secondary Planck-compatible reference C2 with $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (a separately implemented sensitivity check sharing the same Planck sky, not an independent confirmation). Within the predeclared bridge family, constrained-domain predictive validation (SN plus DESI $z \leq 1.8$ folds; Ly α scored separately as a failed continuation test) selects a compact five-coordinate order-3 map of the branch difference with local projected Jacobian rank 5 of 5 fitted coordinates ($\chi^2 = 1459.96$, only 8.04 above the 21-coordinate control). The frozen $m3$ map improves held-out supernova prediction by $\Delta\chi^2 = -76.1$ against the bare null (bare ΛCDM , zero fitted bridge coordinates) and improves constrained DESI ($z \leq 1.8$) by $\Delta\chi^2 = -6.90$ (bare $\chi^2 = 18.08$ against $m3$ $\chi^2 = 11.17$); the separate $z = 2.33$ Ly α continuation test fails strongly, and the matched constrained-domain null bootstrap (250 replicates; same null/candidate family, data split, bounds and selection rule) gives $P(\text{select} \geq m3) = 0.216$ with Ly α catastrophe fraction 0.788 recorded separately; the bootstrap tests the complexity of the selected representation, not the existence of the numerical difference between frozen endpoint histories, so order selection alone is not evidence of a discovery. Zero ruler-clock split is allowed ($\Delta\chi^2(\epsilon=0) = 0.124$). The reference-sensitivity check shows the two R–CMB difference fields share one dominant direction (cosine 0.9997, principal angle 1.5° , common-mode fraction ≥ 0.998 per interval, largest amplitude near $z \approx 0.65$), while across the constrained bins the C1–C2 RMS difference is about 1–8% of the R–CMB difference (0.037 and 0.013 RMS ratios on the constrained and high intervals). These results characterize a reproducible phenomenological difference between the frozen endpoint histories over $z \leq 1.8$; they do not establish its dynamical origin or resolve the Hubble tension. In short: the distance sector prefers systematically higher expansion than the Planck references in a redshift-dependent way that a compact map summarizes but does not explain, and the map does not extend to Ly α .

1 Introduction

The Hubble tension is an observational disagreement between the local supernova distance ladder, $H_0 = 73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [1], and the Planck 2018 TT/TE/EE spectrum, $67.4 \pm$

$0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ in flat Λ CDM [2], reviewed in Verde et al. [3] and Knox and Millea [4]. The mismatch extends beyond a single number: DESI DR2 BAO [5] sharpens the late-time distance–redshift relation while the Pantheon+SH0ES compilation [6, 7] maps the supernova expansion history, and neither a distance-calibrated nor a CMB-calibrated history fits all sectors at once. Observed values, uncertainties, and covariance matrices are held fixed across model comparisons; only theory predictions change between branches.

The standard question this paper asks is: is the disagreement reducible to a compact redshift-dependent difference between independently calibrated background histories? Rather than fitting another cosmological model with additional dynamical freedom, we test whether the observable gap between already-calibrated endpoints has low-dimensional, predictable structure.

The experimental design uses three frozen endpoints. The distance-compatible history R reaches $H_0 = 72.80 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and fits DESI DR2 BAO and Pantheon+SH0ES supernovae. The CMB-compatible history C1 is a Planck TT/TE/EE-compatible reference with $H_0 = 67.99 \text{ km s}^{-1} \text{ Mpc}^{-1}$ produced with the stated CAMB/likelihood configuration. The secondary Planck-compatible reference C2 is a separately implemented baseline with $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, built through its own likelihood and spectral-parity path for reference-sensitivity checking. C1 and C2 are separately implemented in pipeline/model space, not statistically independent in data space: they share the Planck sky but differ in construction pipeline, parameter choices, and calibration route.

This differs from fitting another cosmological model in four ways. First, the endpoints are constructed separately, before any comparison is made. Second, bridge dimensionality is selected with held-out prediction, not training χ^2 . Third, higher flexibility is allowed to fail: overfitting orders and boundary-limited families are retained as frozen adversarial controls. Fourth, the reference-branch robustness test changes the CMB reference branch without changing the hypothesis: the same compact map must describe both Δ_1 and Δ_2 .

The headline result, over the constrained domain $z \leq 1.8$, is

$$\Delta_1(z) \simeq \Delta_2(z), \quad \|\Delta_C\| \ll \|\Delta_{1,2}\|,$$

where the fields $\Delta_1(z) \equiv \ln H_R(z) - \ln H_{C_1}(z)$, $\Delta_2(z) \equiv \ln H_R(z) - \ln H_{C_2}(z)$, and $\Delta_C(z) \equiv \ln H_{C_1}(z) - \ln H_{C_2}(z)$ are defined in Section 2.4 and summarized in the nomenclature table (Table 1). The two R–CMB difference fields share one dominant direction (cosine 0.9997, principal angle 1.5°), peak near $z \approx 0.65$, and transfer across branches with fractional residual degradation 0.068, whereas the CMB–CMB difference is much smaller throughout (about 1–8% of the R–CMB difference across the constrained bins).

In plain terms, the distance-sector endpoint requires a systematically higher expansion rate than the Planck-calibrated references, and the size of that mismatch changes with redshift rather than behaving like a simple constant H_0 offset. A compact five-parameter map summarizes that pattern over $z \leq 1.8$, but its statistical significance and physical origin are not established, and the same compact extrapolation fails at the Ly α point at $z = 2.33$.

This paper identifies and characterizes that empirical deformation. The principal result of this work is therefore not a specific dynamical model, but an empirically defined expansion-history target that candidate explanations of the CMB–distance discrepancy can be tested against. Its physical origin remains deliberately unaddressed. Section 2 defines the data and the R, C1, and C2 endpoints (with the nomenclature table immediately below); Section 3 the empirical reconstruction method; Section 4 the predictive model-selection tests; Section 5 the reference-branch robustness test; Section 6 the results; and Sections 7–9 the interpretation, limitations, and conclusions.

2 Data and endpoint definitions

The three expansion histories compared in this paper were obtained separately, before any comparison was made. Each is frozen: no endpoint is refit, retuned, or extended anywhere in this analysis. All headline values resolve through generated macros to the frozen artifacts cited below. Symbols are defined once in the nomenclature table (Table 1); throughout the main text, R is the distance-compatible endpoint, $C1$ and $C2$ are the two Planck-compatible endpoints, and $m3$ is the selected compact reconstruction basis.

Table 1: Nomenclature: symbols used in the main analysis.

Symbol	Meaning	How obtained	Used for
R	distance-compatible back-ground	frozen distance-sector end-point	reference deformation
C_1	Planck-compatible end-point 1	Planck TT/TE/EE calibrated CAMB reference	CMB reference
C_2	Planck-compatible end-point 2	secondary Planck/CAMB baseline	robustness reference
Δ_1	$\ln H_R(z) - \ln H_{C_1}(z)$	derived	comparison
Δ_2	$\ln H_R(z) - \ln H_{C_2}(z)$	derived	comparison
Δ_C	$\ln H_{C_1}(z) - \ln H_{C_2}(z)$	derived	reference spread
K_H	overall amplitude of the bridge map	fitted (bridge coordinate)	compact representation
c_j	Legendre shape coefficients ($j = 0, 1, 2$)	fitted (bridge coordinates)	compact representation
$m3$	Legendre order-3 reconstruction	CV-selected	compact representation
ϵ	reciprocity parameter	profiled	diagnostic only

2.1 Distance-compatible endpoint R

R is a frozen phenomenological low-redshift expansion history obtained from the distance-sector optimization, with $H_0 = 72.80 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $r_d = 135.39 \text{ Mpc}$; it is not presented as a complete early-universe cosmology. It describes what the fitted low-redshift distance sector prefers; no viable recombination, BBN, or perturbation completion is claimed. The compressed distance prior [8] used during endpoint discovery was validated only for conventional late-time models; it did not validate the $r_d = 135.39 \text{ Mpc}$ early-universe physics.

Concretely, R is the smooth low-redshift control history (`runs/cmb_bridge/r3/smooth_lowz_history.npz`, pinned in `paper/evidence/cmb_bridge_r6/endpoint_manifest.json`), a direct- $H(z)$ spline control carrying no lapse degree of freedom. It reaches $H_0 = 72.80 \text{ km s}^{-1} \text{ Mpc}^{-1}$ with $r_d = 135.39 \text{ Mpc}$ and fits the DESI DR2 and Pantheon+SH0ES distance sector. Nothing in Sections 4 or 5 depends on the history of how R was found.

2.2 CMB-compatible endpoint $C1$

$C1$ is a Planck TT/TE/EE-compatible reference background with $H_0 = 67.99 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $r_{\text{drag}} = 147.15 \text{ Mpc}$, produced with the stated CAMB/likelihood configuration (`runs/cmb_bridge/`

r4/bare_background.npz). No distance-branch parameter, table, or likelihood entered its construction.

2.3 Secondary Planck-compatible reference C2

C2 is a secondary Planck-compatible Λ CDM reference with $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, used only as a reference-sensitivity check, evaluated through its own Planck likelihood (`runs/cmb_branch/baseline/lcdm_table.npz`). It shares the Planck sky with C1 and uses nearly the same Λ CDM background family with its own parameter choices and calibration route, so it cannot provide independent cosmological confirmation. C1 and C2 are separately implemented in pipeline/model space, not statistically independent in data space; both use Planck information. Table 3 compares the two constructions row by row.

2.4 Common observable representation

All three endpoints are compared on one common uniform- u grid of 2000 nodes spanning $0 \leq z \leq 2.33$ (`paper/evidence/cmb_bridge_r6/endpoint_redshift.json`). The difference fields

$$\Delta_1(z) \equiv \ln H_R(z) - \ln H_{C_1}(z), \quad \Delta_2(z) \equiv \ln H_R(z) - \ln H_{C_2}(z), \quad \Delta_C(z) \equiv \ln H_{C_1}(z) - \ln H_{C_2}(z)$$

are evaluated on that grid and frozen in `paper/evidence/cmb_bridge_r6/difference_fields.npz`. The constrained domain is $z \leq 1.8$, covering all non- $\text{Ly}\alpha$ DESI blocks; the $\text{Ly}\alpha$ anchor at $z = 2.33$ tests the continuation law, not the low-redshift deformation (`paper/evidence/cmb_bridge_r6/domain_split.json`).

2.5 Branch acoustic calibration

Each endpoint carries its own acoustic ruler, traced to frozen artifacts in the branch calibration contract (`paper/evidence/branch_calibration_contract.json`). Boltzmann rulers are computed with CAMB [9]:

Table 2: Branch acoustic calibration: every endpoint’s ruler. The compressed distance prior [8] did not validate nonstandard early-universe physics for R.

Endpoint	r_d [Mpc]	r_s [Mpc]	Fixed or recomputed?	Source
R	135.39	— (not used)	fixed (pinned control fit)	control fit
C1	147.15	144.50	recomputed (CAMB stage-0)	stage-0 background
C2	147.11	144.45	recomputed (CAMB baseline)	baseline acoustic

R uses an early-ruler value different from C1/C2; this ruler difference is part of the measured branch separation, not a shared calibration. The R ruler is roughly 0.92 of either CMB ruler, an $\sim 8\%$ smaller ruler. This is a property of the phenomenological distance endpoint, not a demonstrated physical sound horizon: R’s $r_d = 135.39$ Mpc is a phenomenological endpoint value and is not validated as early-universe physics. The compressed Planck distance prior [8] was validated only for conventional Λ CDM, w CDM, and CPL models and cannot by itself validate such nonstandard pre-recombination physics; no BBN viability is claimed (no BBN fit is performed). The BAO forward model is branch-specific:

$$D_M/r_d, \quad D_H/r_d, \quad D_V/r_d, \quad (1)$$

evaluated with each branch’s own ruler from Table 2.

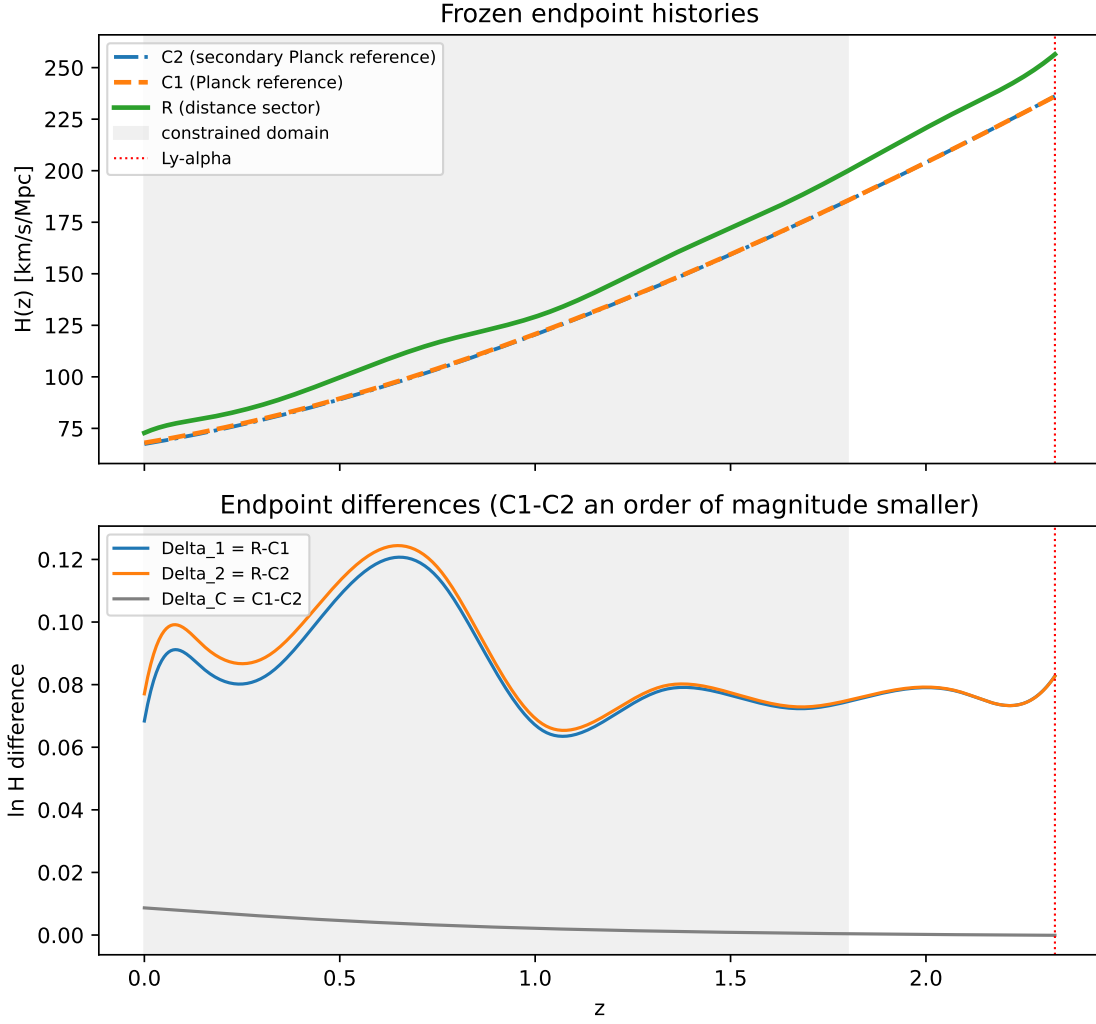


Figure 1: Frozen endpoint overview (no fits in this figure). Top: the three frozen histories H_R, H_{C_1}, H_{C_2} ; C1 and C2 nearly overlap in the upper panel and distinct line styles are used to keep both visible. Bottom: their logarithmic differences $\Delta_1, \Delta_2, \Delta_C$; across the constrained bins the C1–C2 RMS difference is about 1–8% of the R–CMB difference. Shading marks the constrained domain $z \leq 1.8$; the dotted line marks the Ly α continuation test at $z = 2.33$.

Table 3: Endpoint construction comparison: C1 versus C2.

	C1	C2
Underlying CMB observations	Planck TT/TE/EE	Planck TT/TE/EE
Boltzmann code	CAMB 2.0.4	CAMB (baseline pipeline)
Likelihood implementation	Planck TT/TE/EE likelihood (CAMB-based)	stock baseline Planck likelihood
Cosmological parameterization	flat Λ CDM stage-0	flat Λ CDM
Primordial spectrum parameterization	power-law tilt	power-law tilt, $n_s = 0.965$
Optimizer/sampler	background tuning (stage-0)	fixed-parameter baseline with spectral parity
Background extraction	frozen CAMB table	CAMB-sampled $H(z)$ table
H_0 [km s ⁻¹ Mpc ⁻¹]	67.99	67.4
Ω_m	0.3075	0.3134
r_d [Mpc]	147.15	147.11
r_s [Mpc]	144.50	144.45

C1 and C2 are separately implemented in pipeline/model space, not statistically independent in data space; both use Planck information.

Observed values, uncertainties, and covariance matrices are held fixed across model comparisons. Only theory predictions change. What follows groups the likelihoods into the low-redshift distance sector, the high-redshift CMB sector, and the controls that keep the comparison honest.

2.6 Low-redshift distances

DESI DR2 BAO [5] enters through the collaboration distance ratios D_M/r_d , D_H/r_d , D_V/r_d with the published covariances; DESI values are fixed and only the theory prediction changes between branches.

The supernova block is the full Pantheon+SH0ES compilation [1, 6, 7] as one 1657-row covariance likelihood with one common profiled absolute magnitude M (row-selection contract, `paper/evidence/sn_selection_contract.json`: 1701 raw rows selected by the exact mask (`zHD > 0.01`) | `IS_CALIBRATOR` to 1657 rows, of which 77 calibrators and 1580 Hubble-flow rows). With $\mu_{\text{no } M}$ the branch distance modulus at $M = 0$,

$$r_0 = m_B^{\text{corr}} - \mu_{\text{no } M}, \quad (2)$$

$$\hat{M} = \frac{\mathbf{1}^T C^{-1} r_0}{\mathbf{1}^T C^{-1} \mathbf{1}}, \quad (3)$$

$$\chi_{\text{joint}}^2 = (r_0 - \hat{M} \mathbf{1})^T C^{-1} (r_0 - \hat{M} \mathbf{1}), \quad (4)$$

where C is the full 1657×1657 statistical-plus-systematics covariance. The primary joint likelihood profiles one common M (Eq. (3)). Ladder and non-ladder subsets are separately profiled for diagnostics,

$$\hat{M}_{\text{ladder}}, \quad \hat{M}_{\text{nonladder}}, \quad (5)$$

each using its marginal covariance with an independent M , so $\chi_{\text{ladder}}^2 + \chi_{\text{non-ladder}}^2 \neq \chi_{\text{joint}}^2$. These subset values are audit diagnostics (diagnostic-only) and are not additive contributions to the primary joint likelihood.

Calibrator rows carry the published Cepheid host distance $\mu_{\text{cal}} = \text{CEPH_DIST}$, on the SH0ES Cepheid period–luminosity calibration [1, 10]. Under the adopted calibration contract, no additional

theory-side correction is applied to local calibrator distance moduli: $\Delta\mu_{\text{cal}} = 0$ (validation disposition *calibrator-invariance-assumption-only*). The early field is negligible locally ($|g_E(0)| \sim 3.5 \times 10^{-11}$) and same-epoch factors cancel qualitatively, but the local-propagation bound 4.6×10^{-4} mag exceeds the 10^{-4} mag target, so the assumption is supported qualitatively, not quantitatively established. The luminosity-distance construction $D_L = (1+z)D_M$ is Etherington’s reciprocity result [11] and assumes massless photons, null geodesics, and photon-number conservation. Hubble-flow rows use z_{HD} for the transverse integral and the $z_{\text{HD}}/z_{\text{HEL}}$ prefactor $D_L = (1+z_{\text{HEL}})D_M(z_{\text{HD}})$.

2.7 High-redshift CMB

Two CMB uses appear in this paper and must not be confused. The distance endpoint R carries the phenomenological value $r_d = 135.39$ Mpc, which is not validated as early-universe physics: the compressed Planck distance prior [8] was validated only for conventional Λ CDM, w CDM, and CPL models and cannot by itself establish the viability of nonstandard pre-recombination physics. The C1 and C2 reference constructions are calibrated instead to the full Planck TT/TE/EE spectra with foreground-marginalized likelihoods, evaluated through a Boltzmann solver at each sampled cosmology, and carry full Planck-compatible CAMB rulers near 147.1 Mpc. Sound-speed and drag-epoch conventions follow the standard baryon-loading treatment [12, 13].

2.8 Controls

Three controls keep the branch comparison honest. First, Λ CDM parity: the Boltzmann pipeline reproduces reference Λ CDM spectra before either branch is evaluated, so pipeline error cannot masquerade as branch separation. Second, frozen endpoint artifacts are hash-bound: every number in Sections 6–5 is generated from frozen machine-readable evidence with source-hash binding (`paper/evidence_manifest.yaml`); generation fails on hash mismatch. Headline numerical results are generated from or hash-bound to frozen evidence.

3 Empirical reconstruction method

This paper fits no cosmological model. Its only fitted objects are the bridge coordinates mapping the frozen endpoint histories onto each other over the constrained domain $z \leq 1.8$.

3.1 Bridge coordinates and difference fields

The fitted bridge coordinates at order m are the overall amplitude K_H , the reciprocity parameter ϵ , and the Legendre shape coefficients $c_0, \dots, c_{m-1}$ (at the selected $m = 3$: five fitted coordinates). The compared fields are $\Delta_1(z) \equiv \ln H_R(z) - \ln H_{C_1}(z)$, $\Delta_2(z) \equiv \ln H_R(z) - \ln H_{C_2}(z)$, and $\Delta_C(z) \equiv \ln H_{C_1}(z) - \ln H_{C_2}(z)$ (Section 2.4).

3.2 Likelihood residual vector and SN nuisance profile

The residual vector has 1670 rows: DESI BAO rows plus 1657 Pantheon+SH0ES rows (row-selection contract, `paper/evidence/sn_selection_contract.json`). DESI rows are whitened with the Cholesky factor of the full published covariance; SN rows are whitened with the full 1657×1657 covariance and projected to remove the whitened-ones direction, which analytically profiles the single common absolute magnitude M (Eqs. (2)–(4)). No H_0 target, penalty, or reweighting enters anywhere.

3.3 Fold definitions and constrained selection rule

Order selection uses held-out behavior on identical folds: five supernova conditional folds and six DESI leave-block-out folds at $z \leq 1.8$. The Ly α block at $z = 2.33$ is scored separately as an out-of-domain continuation test and never participates in selection. The predeclared rule selects the smallest m within 2.0 of the best combined held-out score (constrained re-score `constrained_cv_rescore.json` in paper evidence).

3.4 Projected Jacobian rank

Let $r \in \mathbb{R}^{1670}$ be the whitened, nuisance-projected residual vector and $p = (\log K_H, \epsilon, c_0, c_1, c_2)$ the fitted coordinates. The projected whitened Jacobian is

$$J_{ij} = \frac{\partial r_i}{\partial p_j} s_j, \quad (6)$$

with finite-difference steps $h_j = 10^{-6} \max(1, |p_j|)$ and column scales $s = (0.4055, 0.20, 1, 1, 1)$. Its singular values are $[602.38, 157.31, 102.07, 32.03, 22.31]$ (condition number ≈ 27), giving local projected Jacobian rank 5/5 fitted coordinates (rank-operator contract `rank_operator_contract.json` in paper evidence). These are fitted coordinates, not measured observables.

3.5 Bootstrap and information criteria

A parametric bootstrap of the entire selection pipeline under the bare null (null contract `null_model_contract.json` in paper evidence) calibrates order-selection frequency; it tests the complexity of the selected representation, not the existence of the numerical difference between frozen endpoint histories. Information criteria are reported descriptively (parsimony penalty only), never as detections.

Figures and tables are generated exclusively from frozen evidence, all SHA-pinned in the manifest `paper/evidence_manifest.yaml`; generation fails if a source hash mismatches. All reported values are best fits; no credible or confidence intervals are claimed, and every comparison is descriptive with no Wilks-theorem calibration.

4 Predictive model-selection tests

This section reconstructs the difference between the distance-compatible history R and the CMB-compatible history C1 with a compact coordinate representation. All bridge numbers resolve to frozen evidence in `paper/evidence/cmb_bridge_r4v/` through registry macros; headline numerical results are generated from or hash-bound to frozen evidence.

4.1 Model-selection procedure

Within the predeclared bridge family (shifted-Legendre shape basis with split ruler/clock calibration) and fixed likelihood contract, orders $m = 2-6$ were fitted from multiple optimizer starts with a boundary audit, a rank audit, and epsilon profiling. Constrained-domain order selection uses held-out behavior on identical folds over $z \leq 1.8$: five supernova conditional folds and six DESI leave-block-out folds at $z \leq 1.8$. The Ly α block at $z = 2.33$ is scored separately as an out-of-domain continuation test and never participates in selection (constrained re-score `constrained_cv_rescore.json` in paper evidence; no refitting — a re-score of frozen folds only).

4.2 Main result

Constrained predictive validation selects shifted-Legendre order 3 ($m = 3$) with five fitted coordinates: total $\chi^2 = 1459.96$, only 8.04 above the 21-coordinate control despite 16 fewer fitted coordinates, with local projected Jacobian rank 5 of 5 fitted coordinates after covariance whitening and nuisance projection (Section 3). Scores are tabulated with both columns:

Table 4: Constrained-domain held-out scores by order (selection column) with the Ly α continuation score alongside. The star marks the selected order under the predeclared smallest- m -within-2.0 rule. Frozen folds, re-scored; no refitting.

m	SN held-out	DESI $z \leq 1.8$ held-out	combined (selection)	DESI Ly α (continuation)
2	8201.1	12.798	8213.920	33.6
3 ★	8153.1	11.174	8164.276	59.7
4	8151.9	12.252	8164.149	262.5
5	8166.8	20.010	8186.770	5448.6
6	8160.9	23.091	8184.005	19969.2

The raw constrained minimum is $m = 4$ (8164.15 against 8164.28 at $m = 3$), a margin of about 0.13 that does not overturn the predeclared smallest- m rule: $m = 3$ remains selected. Selection therefore does not depend on the Ly α catastrophe. Information criteria are reported descriptively, side-by-side: $k = 5$, $\chi^2 = 1459.96$, $\Delta\chi^2_{\text{control}} = 8.04$, AIC ≈ 1469.96 against ≈ 1493.92 for the $k = 21$ control ($\Delta\text{AIC} = -23.96$). In words: $m3$ gives up 8.04 in χ^2 while using 16 fewer fitted bridge coordinates; AIC therefore favors the compact representation primarily through its parsimony penalty. This is not interpreted as a detection or a calibrated probability. The measurable shape is concentrated at low to intermediate redshift (DESI improvement peak $0.4 < z < 0.8$; redshift audit in evidence). The Legendre representation is a compression basis for the empirical deformation, not a proposed fundamental expansion law.

4.3 Predictive validation and the failed continuation

Higher orders degrade constrained held-out prediction (Table 4). The bare null is the bare ΛCDM endpoint with zero fitted bridge coordinates (K_H fixed to no deformation, ϵ not applicable) and a single common M profiled per fold-train (null contract, `paper/evidence/cmb_bridge_r5/null_model_contract.json`). Against this null on identical folds, the frozen $m = 3$ map improves held-out supernova prediction by $\Delta\chi^2 = -76.1$ and improves constrained DESI ($z \leq 1.8$) by $\Delta\chi^2 = -6.90$ (bare $\chi^2 = 18.08$ against $m3$ $\chi^2 = 11.17$); the separate $z = 2.33$ Ly α continuation test fails strongly. The Ly α block is then scored as an out-of-domain continuation test, where every order fails badly (Ly α column of Table 4). The empirical continuation fails, and therefore the reconstruction’s validated domain ends at $z \leq 1.8$; this bounds the claim rather than rescuing the model.

4.4 Prosecution: selection is not significance

A parametric bootstrap of the entire constrained-domain selection pipeline under the bare null (250 replicates; same null/candidate family, data split, bounds and selection rule; constrained bootstrap `selection_bootstrap_constrained.json` in paper evidence) gives $P(\text{select} \geq m3 \mid \text{null}) = 0.216$, reported exactly as obtained with no threshold tuning. The Ly α catastrophe statistic is recorded

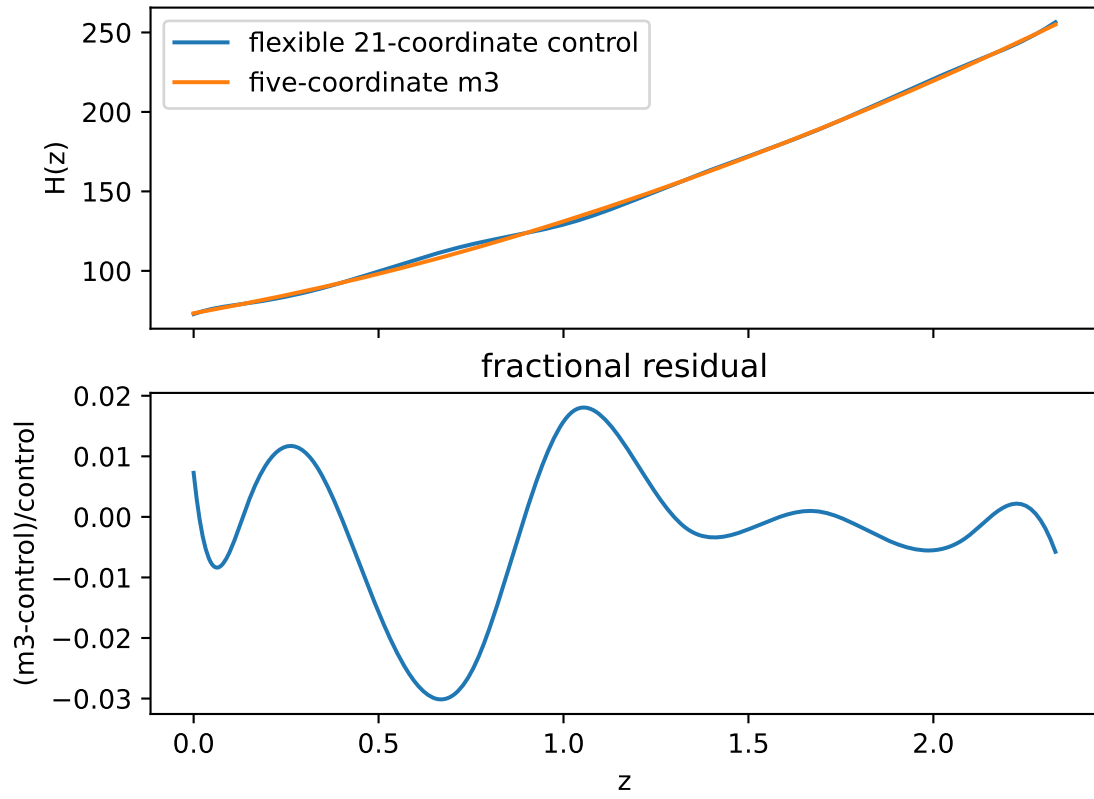


Figure 2: Five-coordinate m3 reconstruction compared with the flexible 21-coordinate control. The lower panel shows $(H_{m3} - H_{\text{control}})/H_{\text{control}}$.

separately and never chooses m . This bootstrap tests the complexity of the selected representation, not the existence of the numerical difference between frozen endpoint histories. Model-order selection alone is not evidence of a discovery. The reference-sensitivity analysis is a separate robustness check, not a significance test. The result is presented as a compact reconstruction with demonstrated held-out behavior, not evidence of a discovery.

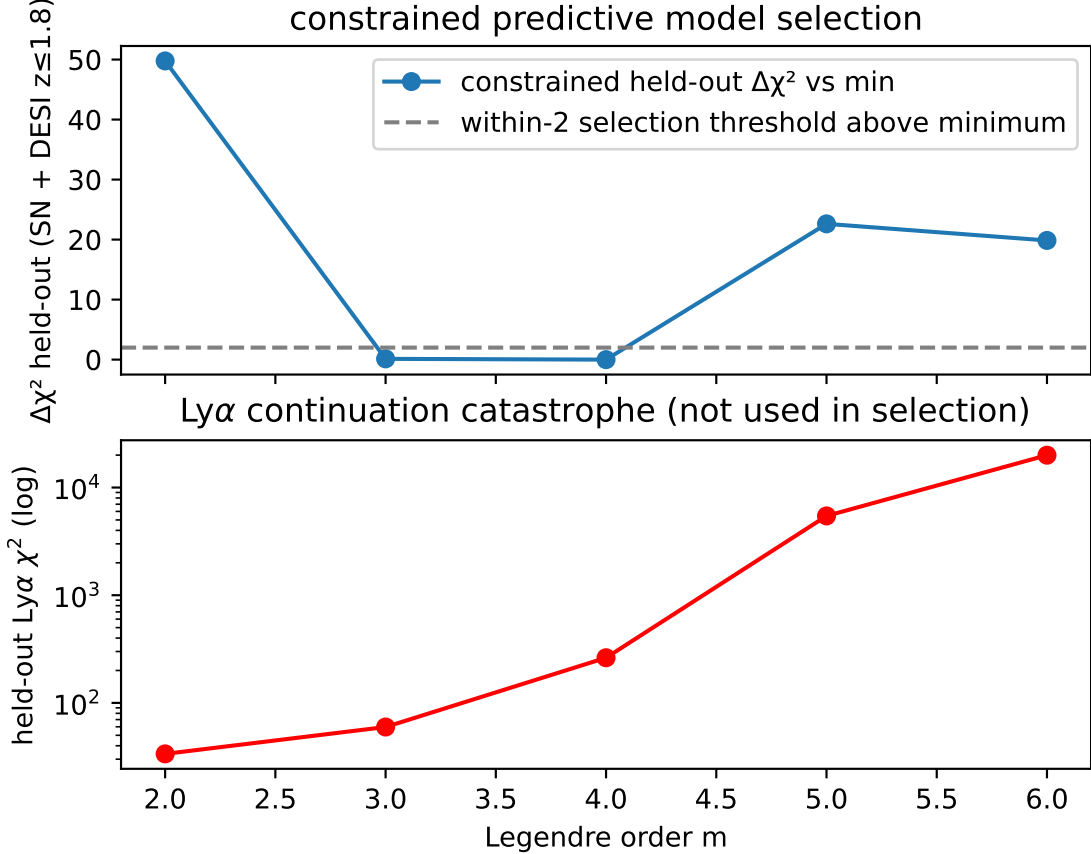


Figure 3: Constrained predictive model selection: held-out $\Delta\chi^2$ versus the constrained minimum (top; SN plus DESI $z \leq 1.8$), with the $\text{Ly}\alpha$ continuation catastrophe on a logarithmic scale (bottom; not used in selection). The $m3/m4$ near-tie is visible; the dashed line is the within-2 selection threshold above the minimum, and the predeclared smallest- m -within-2.0 rule selects $m3$.

4.5 Reciprocity test

The ruler/clock split is consistent with zero: $\Delta\chi^2(\epsilon=0) = 0.124$, with 1-sigma profile approximately $[-0.012, +0.005]$. No nonreciprocal calibration is detected; ϵ is not detected.

4.6 Why increased flexibility is rejected

The analysis actively rejected better-looking training solutions. The roughness family reaches $\chi^2 = 1453.34$ but its leading shape coefficient sits exactly at the imposed bound ($c_1 = +0.5$) with the local descent gradient pointing outward, so its optimum depends on the box: it is boundary-limited and retained as a supporting basis only. The Legendre $m = 6$ map reaches good training $\chi^2 = 1455.19$

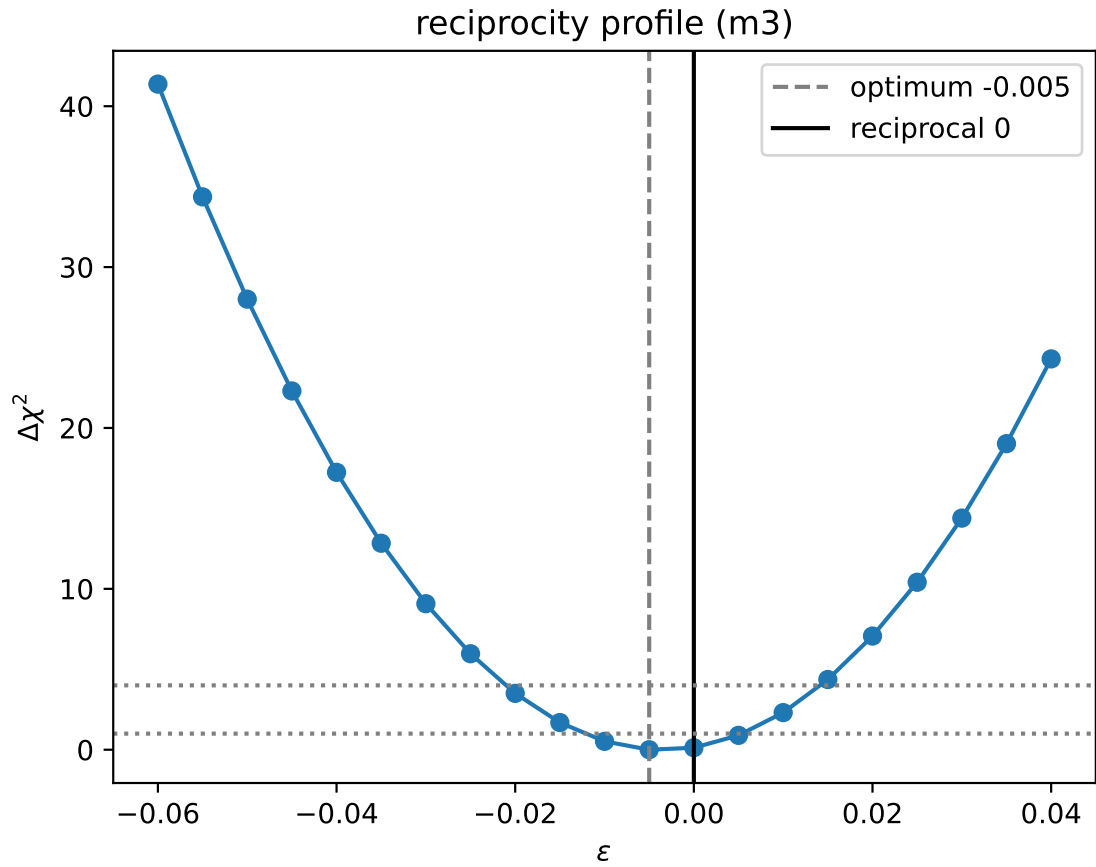


Figure 4: Reciprocity profile $\Delta\chi^2(\epsilon)$ in the selected $m = 3$ map: optimum near -0.005 and zero at $\Delta\chi^2 = 0.124$, with horizontal $\Delta\chi^2 = 1$ and 4 levels.

with an interior optimum but fails the held-out Lyman-alpha continuation test (Table 4); it is therefore not promoted. Both are retained as frozen adversarial controls in evidence.

4.7 Scope of this section

The analysis establishes an empirical background reconstruction, not a fundamental field theory. Whether this map is the observable limit of new dynamics remains unresolved. Chronometer, growth, lensing, time-delay, and siren sectors are not predicted without a covariant completion; the sector table in evidence records every status as NOT_PREDICTED_WITHOUT_COMPLETION. The *m3* representation is not interpreted as a fundamental polynomial law.

5 Reference-branch robustness

This is the paper’s centerpiece. The reconstruction above is measured against one particular CMB-compatible history C1. This section repeats the comparison against a second, separately implemented CMB-compatible history C2 and shows the deformation is shared. All headline metrics resolve from frozen evidence JSON through registry macros, never handwritten.

5.1 Endpoint invariance

Define the logarithmic difference fields on the common grid (Section 2.4):

$$\Delta_1(z) = \ln H_R(z) - \ln H_{C1}(z), \quad (7)$$

$$\Delta_2(z) = \ln H_R(z) - \ln H_{C2}(z), \quad (8)$$

$$\Delta_C(z) = \ln H_{C1}(z) - \ln H_{C2}(z), \quad (9)$$

with the common and disagreement combinations

$$\Delta_{\text{common}}(z) = \frac{\Delta_1(z) + \Delta_2(z)}{2}, \quad (10)$$

$$\Delta_{\text{disagreement}}(z) = \frac{\Delta_1(z) - \Delta_2(z)}{2}. \quad (11)$$

The quoted common-mode fraction is

$$f_{\text{common}} = 1 - \left[\frac{\text{RMS}(\Delta_{\text{disagreement}})}{\text{RMS}(\Delta_{\text{common}})} \right]^2, \quad (12)$$

evaluated per redshift interval on the constrained domain. Changing the CMB reference branch from C1 to C2 does not change the hypothesis: the same compact map must describe both separations.

5.2 Common-mode decomposition

On the constrained domain ($z \leq 1.8$), the two R-CMB fields lie substantially on top of one another. The common-mode fraction exceeds 0.998 in every redshift interval, and the two-vector singular decomposition is dominated by a single shared direction (dominance ratio 76.3). The two fields peak at the same location, near $z \approx 0.65$, while across the constrained bins the CMB-CMB RMS difference is about 1–8% of the R-CMB difference (RMS ratios 0.037 on the peak interval and 0.013 at high redshift).

5.3 Peak location and data support

The fitted difference reaches its largest amplitude near $z \approx 0.65$. Because this region also carries substantial observational leverage, we do not interpret the peak redshift as a fundamental physical scale. A bounded diagnostic (peak-support summary in paper evidence, no refitting) compares the frozen peak against Pantheon+ SN count and inverse-variance densities, DESI BAO information density and block locations, and the already-computed fit-improvement attribution: the peak does not coincide with any simple data-density maximum. The recorded disposition is peak-not-explained-by-simple-data-density. R-C1 and R-C2 both peak near 0.65; the frozen m3 curve (maximum near 0.39) captures their broad low/intermediate- z trend but not their exact peak. The peak’s origin remains unresolved.

5.4 Cross-branch transfer

The frozen order-3 basis transfers across branches: shape amplitudes refit independently agree, and parameters learned on one CMB branch predict the other separation with residual degradation fraction 0.068 ($\leq 6.8\%$ cross-branch). One dominant shared direction carries the deformation; the residual branch-specific component is small.

5.5 Statistical status of the reference-branch robustness test

The m3 order-selection bootstrap tests model complexity; it does not test reference-construction similarity. The reference-sensitivity metrics quantify robustness of frozen best-fit histories: how much the measured R-CMB separation moves when the CMB reference construction changes. C1 and C2 are not statistically independent (both use Planck information; Section 2.3), and the two difference fields share the same R endpoint ($\Delta_1 = R - C_1$, $\Delta_2 = R - C_2$), so their similarity is mathematically correlated. Endpoint posterior uncertainty was not propagated. Therefore no frequentist p-value is assigned to the cosine or common-mode statistic. The reference-branch robustness test shows a shared low-dimensional deformation across the tested constructions; it does not confirm a theory.

5.6 Constrained domain and Ly α continuation

All reference-sensitivity metrics above are quoted on the constrained domain $z \leq 1.8$, covering every non-Ly α DESI block. The Ly α anchor at $z = 2.33$ is a continuation test of the reconstruction law, not a measurement of the low-redshift deformation: flexible extrapolators generically fail there, under the null as on the data (Ly α catastrophe fraction 0.788 in the matched constrained-domain bootstrap, recorded separately and never used in selection).

5.7 Comparison with the fit-improvement attribution

The deformation RMS (peak bin $0.4 < z < 0.8$) and the fit-improvement attribution (peak bin $0 < z < 0.4$) both concentrate below $z \sim 0.8$, though their peak bins differ; high-redshift behavior is unconstrained in both. The reference-sensitivity check therefore supports, rather than replaces, the compact reconstruction: the same five-coordinate map describes a deformation that is reference-insensitive within the two tested Planck-compatible constructions.

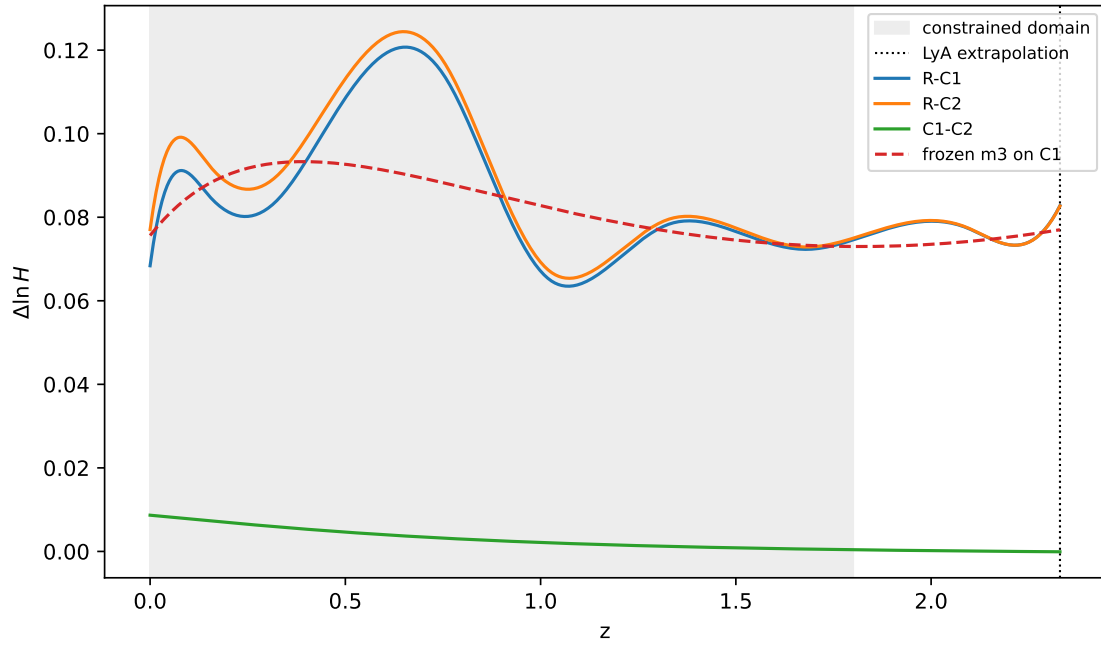


Figure 5: Reference-sensitivity overlay: the two R-CMB difference fields lie on top of one another across the constrained domain (R-C1 and R-C2 nearly coincide; C1-C2 is much smaller), and the frozen m3 reconstruction captures the broad low/intermediate-redshift trend but not the exact peak near $z \simeq 0.65$. Shading marks the constrained domain; the dotted line marks the Ly α extrapolation test.

6 Results

This section states the paper’s empirical results in one place. Construction details are in Sections 2–5. Headline numerical results are generated from or hash-bound to frozen evidence.

6.1 Headline branch separation

The phenomenological low-redshift distance-sector endpoint R ($H_0 = 72.80 \text{ km s}^{-1} \text{ Mpc}^{-1}$; not presented as a complete cosmological model) differs substantially from both Planck-compatible references C1 ($H_0 = 67.99 \text{ km s}^{-1} \text{ Mpc}^{-1}$) and C2 ($H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, secondary sensitivity reference) across the constrained domain $z \leq 1.8$. The two R–CMB difference fields Δ_1 and Δ_2 (defined in Section 2.4) share one dominant common mode (common-mode fraction ≥ 0.998 per interval), while the CMB–CMB difference Δ_C is much smaller throughout: across the constrained bins, the C1–C2 RMS difference is about 1–8% of the R–CMB difference (RMS ratios 0.037 and 0.013 on the constrained and high intervals).

6.2 Compact reconstruction

A five-coordinate order-3 (m3) map gives a compact empirical description of the constrained-domain ($z \leq 1.8$) branch difference with local projected Jacobian rank 5 of 5 fitted coordinates ($\chi^2 = 1459.96$, only 8.04 above the 21-coordinate control; AIC ≈ 1469.96 against ≈ 1493.92 for the control, reported descriptively as parsimony rather than evidence). Selection is predictive (held-out supernova and DESI $z \leq 1.8$ folds, Section 4) and carries no significance of its own: the matched constrained-domain null bootstrap tests representation complexity, not existence of the endpoint separation (null bootstrap $P(\text{select} \geq m3) = 0.216$). Zero ruler-clock split is allowed ($\Delta\chi^2(\epsilon=0) = 0.124$). The fitted difference reaches its largest amplitude near $z \approx 0.65$; because this region also carries substantial observational leverage, we do not interpret the peak redshift as a fundamental physical scale (Section 5.3).

6.3 Reference-branch robustness

The frozen m3 basis transfers across the two Planck references with small residual degradation (fraction 0.068): parameters learned on one reference construction predict the other separation. The two R–CMB fields share one dominant direction (cosine 0.9997, principal angle 1.5°). This reference-sensitivity check shows the deformation is insensitive to the tested CMB-reference construction; it does not assign a frequentist p-value to the similarity (Section 5.5). Neither branch fits everything and no tension resolution is claimed. High-redshift continuation through the Lyman-alpha anchor remains unvalidated.

7 Discussion

7.1 What appears robust

Within the tested reconstructions over $z \leq 1.8$, six features recur. First, a common R–CMB deformation: the two R–CMB difference fields lie on top of one another (common-mode fraction ≥ 0.998 per interval). Second, a peak near $z \approx 0.65$ (largest amplitude; not a fundamental scale). Third, low/intermediate- z concentration: both the deformation RMS (peak bin $0.4 < z < 0.8$) and the fit-improvement attribution (peak bin $0 < z < 0.4$) concentrate below $z \sim 0.8$, though their peak bins differ. Fourth, one dominant shared direction (cosine 0.9997, principal angle 1.5° , SVD

dominance 76.3). Fifth, a compact representation: five coordinates suffice ($\chi^2 = 1459.96$, local projected Jacobian rank 5 of 5 fitted coordinates, $\text{AIC} \approx 1469.96$ against ≈ 1493.92 descriptively). Sixth, reference insensitivity across C1/C2: cross-branch transfer degrades by fraction 0.068 or less while the CMB–CMB difference stays much smaller (about 1–8% of the R–CMB difference across the constrained bins).

7.2 What is not established

No physical origin is established. No fundamental metric modification and no modified gravity are demonstrated. R is a phenomenological low-redshift endpoint, not presented as a complete cosmological model: it describes what the fitted low-redshift distance sector prefers, with no claimed recombination, BBN, or perturbation completion. No tension resolution is claimed: neither branch fits everything. No statistical significance is claimed for the representation complexity: the matched constrained-domain null bootstrap gives $P(\text{select} \geq m3) = 0.216$, so order selection alone is not evidence. No high-redshift continuation law is validated: the $\text{Ly}\alpha$ anchor is an extrapolation test that flexible maps generically fail.

7.3 Possible physical interpretations

We list categories, not endorsements. A first category is evolving late-time background dynamics (dark-energy-like histories reproducing the same $H(z)$ deformation). A second is early-universe and ruler physics (recombination or drag-epoch effects shifting the acoustic scale, or combinations of early and late effects). A third is modified gravity and modified propagation (distance-duality or luminosity-distance effects mimicking expansion structure). A fourth is conventional systematics or calibration effects in one or more of the distance, BAO, or CMB sectors.

7.4 What would discriminate interpretations

Untouched sectors would discriminate where background geometry cannot. Independent tests include cosmic chronometers, redshift-space distortions, weak and CMB lensing, time-delay distances, and standard sirens. No fitting in those sectors is performed in this paper.

8 Limitations

R is a phenomenological distance-sector endpoint: it describes what the fitted low-redshift distance sector prefers and is not presented as a complete cosmological model (no viable recombination, BBN, or perturbation completion is claimed). R’s $r_d = 135.39$ Mpc is a phenomenological endpoint value, not validated early-universe physics: the compressed Planck distance prior [8] was validated only against conventional late-time $\Lambda\text{CDM}/w\text{CDM}/\text{CPL}$ models and cannot by itself establish the viability of nonstandard pre-recombination physics. The C1 and C2 reference constructions are instead calibrated directly to the full Planck spectra, with recomputed CAMB rulers near 147.1 Mpc. Acoustic rulers are branch-specific (Table 2). No calibrated significance attaches to the joint difference. All values are best fits; no posterior intervals are claimed, endpoint posterior uncertainty is not propagated, and C1/C2 are not statistically independent constructions (both use Planck information). Late–early degeneracies beyond the explored range are not covered. The calibrator result covers the declared contract only ($\Delta\mu_{\text{cal}} = 0$, a qualitative assumption, not quantitatively established).

Table 5: Limitations and current status.

Issue	Current status
R physical completion	Not established
R ruler BBN/recombination	Not validated
Endpoint posterior errors	Not propagated
C1/C2 statistical independence	No; both use Planck
m3 order significance	Not established, bootstrap $P = 0.216$
Ly α continuation	Failed
Physical origin	Not identified

The branch-separation result is bounded by what the frozen evidence supports. We claim no resolution of the Hubble tension: R is a phenomenological distance-sector endpoint with no viable recombination or BBN completion, and neither the distance nor the CMB reference fits every sector at once. We claim no modified gravity detection and no proof that Λ CDM fails: C1 is a Planck-compatible Λ CDM reference and fits Planck. What is established is only the separation: the distance-sector endpoint and the Planck-compatible references prefer quantitatively different expansion histories (H_0 72.80 vs 67.99, with branch-specific acoustic rulers per Table 2).

8.1 Relation to neighbouring approaches

DESI DR2 BAO remain compatible with flat Λ CDM in isolation, while combinations with CMB and supernovae can favor evolving dark energy of the w_0w_a form [5, 14, 15]. Our bridge is not a dark-energy equation of state: it is a reconstruction of the required observational transformation between two calibrated histories, with dimensionality selected through held-out data rather than by allowing enough function freedom to improve the joint fit. That selection discipline is also what separates this work from unconstrained Gaussian-process or smoothing reconstructions of the expansion history [16] and from generic cosmography [17], which do not penalize flexibility by prediction. Early-dark-energy solutions modify the pre-recombination universe and its perturbations; the bridge assumes no recombination change and carries no perturbation sector. For the broader solution landscape we refer to the community review [18]. This work does not introduce another equation-of-state fit. It compares separately calibrated histories and asks whether their difference is compact and reference-insensitive within the two tested Planck-compatible constructions. This decomposition is complementary to the scale-vs-shape obstruction argument [19]: uniform sound-horizon rescaling changes scale but cannot remove the BAO–SN shape disagreement, and our R endpoint contains both a large ruler shift ($\sim 8\%$) and a redshift-dependent shape difference. We do not claim the ruler rescaling alone solves the tension, nor anything beyond what was actually compared.

9 Conclusion

Over the constrained domain $z \leq 1.8$, the two Planck-compatible references are much closer to each other than either is to R: across the constrained bins the C1–C2 RMS difference is about 1–8% of either R–CMB difference. Changing between these two Planck-compatible references changes the inferred difference only slightly (reference sensitivity, not independent confirmation).

The two R–CMB difference fields over $z \leq 1.8$ are nearly identical: they share one dominant direction (cosine 0.9997, principal angle 1.5°) with cross-branch transfer degradation fraction 0.068 or

less, reaching largest amplitude near $z \sim 0.65$ without that redshift being claimed as a fundamental physical scale.

A five-coordinate $m3$ basis provides a compact empirical description over $z \leq 1.8$ ($\chi^2 = 1459.96$, local projected Jacobian rank 5 of 5 fitted coordinates), but its selection is not itself statistically significant under the null bootstrap ($P(\text{select} \geq m3) = 0.216$), which tests the complexity of the representation rather than the existence of the numerical difference between frozen endpoint histories. Continuation to $\text{Ly}\alpha$ at $z = 2.33$ fails for the compact extrapolator, bounding the validated domain to $z \leq 1.8$.

The physical origin remains open: no dynamics, no metric theory, and no tension resolution are established here. R is a phenomenological low-redshift endpoint, not presented as a complete cosmological model.

The resulting target is the low-redshift distance-sector behavior represented by R over $z \leq 1.8$, together with the requirement that any physical completion preserve the observed CMB acoustic structure; R itself is not asserted to be such a completion.

A Likelihood details and calibration contract

The joint likelihood sums DESI DR2 BAO terms with published covariances, the compressed CMB prior used during endpoint discovery, and the Pantheon+SH0ES joint term. No external H_0 prior enters the likelihood.

Pantheon+SH0ES is one 1657-row covariance likelihood with one common profiled M . With $r_0 = m_B^{\text{corr}} - \mu_{\text{no } M}$,

$$\hat{M} = \frac{\mathbf{1}^T C^{-1} r_0}{\mathbf{1}^T C^{-1} \mathbf{1}}, \quad \chi_{\text{joint}}^2 = (r_0 - \hat{M} \mathbf{1})^T C^{-1} (r_0 - \hat{M} \mathbf{1}), \quad (13)$$

where C is the full 1657×1657 covariance. Ladder and non-ladder blocks are diagnostics: each uses its marginal covariance with an independent M , so $\chi_{\text{ladder}}^2 + \chi_{\text{non-ladder}}^2 \neq \chi_{\text{joint}}^2$.

Calibrator rows use $\mu_{\text{cal}} = \text{CEPH_DIST}$. Under the adopted calibration contract, no additional theory-side correction is applied to local calibrator distance moduli: $\Delta\mu_{\text{cal}} = 0$; this covers the declared contract only and is a qualitative assumption, not quantitatively established. Hubble-flow rows use z_{HD} for D_M and the $z_{\text{HD}}/z_{\text{HEL}}$ prefactor $D_L = (1 + z_{\text{HEL}})D_M(z_{\text{HD}})$, which assumes massless photons, null geodesics, and photon-number conservation [11].

The endpoint-discovery CMB block is the compressed set $(R, \ell_A, \omega_b h^2)$ with numerical values from [8], validated there against conventional late-time models only:

$$D_M(z_*) = c I(z_*) r_d / (100 h_{\text{rd}}), \quad \ell_A = \pi D_M(z_*) / r_s. \quad (14)$$

This is a compressed $R, \ell_A, \omega_b h^2$ prior, not a TT/TE/EE likelihood, and it cannot by itself validate nonstandard pre-recombination physics for R . The C1/C2 reference constructions are calibrated directly to the full Planck spectra (Section 2.5).

A.1 Calibration contract

No telescope or catalog data were rescaled. All changes are in the forward model, not the data: observed redshifts, magnitudes, covariances, masks, and Cepheid distances are untouched.

DESI path: $z_{\text{obs}} \rightarrow z_{\text{bg}} \rightarrow D_M, D_H, D_V \rightarrow D/r_d$ with each endpoint's own ruler from Table 2; observed D/r_d rows and published covariances are unchanged. Pantheon+SH0ES path: 1657 selected rows including 77 calibrators; z_{HD} for cosmological distance, z_{HEL} in the luminosity prefactor, CEPH_DIST for calibrator μ ; one common profiled M with full 1657×1657 covariance in the fit; source-defined ladder split is diagnostic-only with independent M terms and is non-additive under the joint cross-covariance.

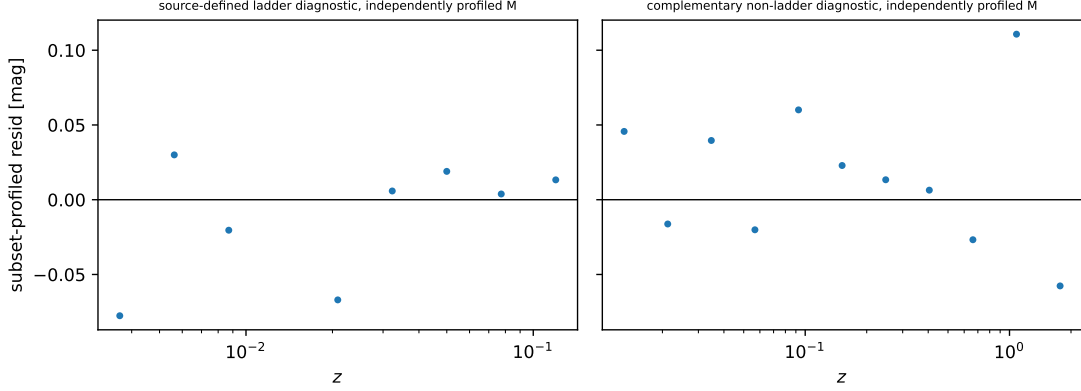


Figure 6: Appendix-only subset-profiled supernova diagnostics: source-defined ladder diagnostic, independently profiled M (left) and complementary non-ladder diagnostic, independently profiled M (right). Non-additive; never used in the joint fit.

B Reproducibility

The driver `scripts/paper/build_expansion_history_paper.py` replays the frozen evidence, rebuilds the results registry, regenerates all figures and tables with source-hash binding, validates claims in release mode, compiles the manuscript with `pdflatex` and `bibtex`, and builds and checks the arXiv bundle. The bundle `paper/dist/arxiv_submission/` is flat and self-contained, excluding run directories, source code, tests, credentials, logs, and caches. Random seeds, code hashes, and data hashes are recorded in the frozen evidence manifests.

Numerical stack. All array arithmetic uses NumPy [20] and all optimization and quadrature use SciPy [21]; versions are pinned in the frozen evidence manifests. Frozen endpoint artifacts and their hashes are listed in `paper/evidence_manifest.yaml`; regeneration fails on hash mismatch, so the published numbers reproduce exactly from the archived evidence.

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